

My Design: Adaptive Technology
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Element I: Testing, Data Collection, and Analysis

We wanted to test our $\frac{3}{4}$ inch screws with our door pieces of $\frac{3}{4}$ inch depth and found no problems. The screws did not splint the wood or come through the other end, probably due to the slight height added by the hinges.

Our shorter roundhead screws were also tested, but didn't work as well as the flathead $\frac{3}{4}$ inch screws that came with the hinges. The roundhead of the longer screws stopped the pieces from folding completely, making it clear which screws we wanted to use. (Screws pictured in *Figure 4*).

We tested two different distances between the wood (*Figure 3: Gap*) when screwing the hinges in when the wood pieces sat next to each other. We started with $\frac{1}{4}$ inch, but faced problems when trying to fold the wooden pieces onto each other, as seen in *Figure 1: No Fold*. The bi-fold door couldn't fold completely.

We separated the wood a little farther at $\frac{3}{8}$ inch and had no problems with fully folding the door, as seen in *Figure 2: Fold*. Because this distance worked, we had no reason to test having the wood next to each other with no gap (or 0 inches).



Figure 1: No Fold

As seen in *Figure 1*, with $\frac{1}{4}$ inch space between the wood for the placement of the hinges, the wood is not able to completely fold onto itself.

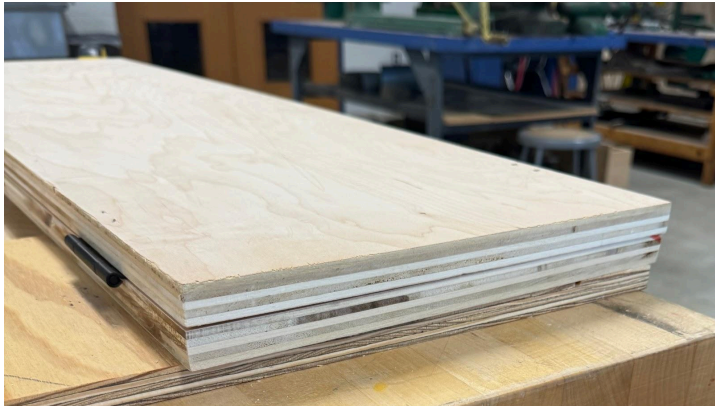


Figure 2: Fold

As seen in *Figure 2*, with $\frac{3}{8}$ inch space between the wood for the placement of the hinges, the wood is able to completely fold onto itself.



Figure 3: Gap

Figure 3 features the gap discussed in the text above this table and is directly related to the results seen in *Figure 1* and *Figure 2*. The gap needs a distance of about $\frac{3}{8}$ inch in order for the wood to fold correctly.



Figure 4: Screws

The screw in *Figure 4* on the left is shorter but has a roundhead while the screw on the right is longer and has a flathead.

By mid January we made a shift in the project to try and hasten the project's completion and we dropped the tactile board and covers for the cubbies which changed a lot of the goals we had with the design and its requirements. We focused on ensuring that the highest priority design requirement was met and so we wanted to ensure that we provided a good solution to utilizing the awkward space taken up by the island. Our design also became focused on optimizing the space while also creating a good experience for the teachers in accessing the space. The testing showed us that empty space's measurement was crucial to us understanding how we can make the experience smoother and more accessible(which is one of our high priority design requirements). The importance of making this experience clean is we don't want to burden them with storage as they already are very concerned with managing that. It also allows for more opportunities in engineering new sensory items while still giving attention to the conflicts surrounding the optimal use of the sensory space.